

Palau National Marine Sanctuary

Science and Monitoring Strategy

2024-2034

May 2023



Compiled by:

Louw Claassens

Anuradha Gupta

Eric Hartge

King Sam

Suggested citation:

Claassens, L., Gupta, A., Hartge, E., Sam, K. 2023. Palau National Marine Sanctuary science and monitoring strategy: 2024-2034. Koror, Palau. 56 pp.

This strategy was developed with generous funding support from:

The GEF7 funded project entitled: Strengthening the Palau National Marine Sanctuary for the Conservation and Management of Global Marine Biodiversity and Sustainable Fisheries (2021-2025)



TABLE OF CONTENTS

02 Executive summary

03 Section 1 – Introduction and background

04 1.1 PNMS science and monitoring strategy overview

07 1.2 PNMS science and monitoring background

10 1.3 PNMS site description

21 Section 2 – Science and monitoring

22 2.1 PNMS science and monitoring strategy development

25 2.2 PNMS science and monitoring framework

40 2.3 PNMS science and monitoring prioritisation and indicators

42 2.4 Adaptive management

43 2.5 Science and management

45 Section 3 – References

48 Section 4 – Appendices

49 4.1 PNMS science and monitoring expert workshop attendees

50 4.2 Research permitting guidance and key contact information

52 4.4 Linkages between the PNMS science and monitoring strategy 2024-2034 and the PNMS strategic plan 2023-2028

Executive Summary

In 2015, the Republic of Palau contributed towards global ocean conservation through the demarcation of 80% of its Exclusive Economic Zone (EEZ) as a no-take, fully protected marine protected area (MPA). The Palau National Marine Sanctuary (PNMS) is one of only a few fully pelagic large-scale MPAs in the world and extends Palau's well-established coastal conservation practices to offshore and deep-sea environments while also materializing Palauan's ancestral ties to their open ocean.

This PNMS science and monitoring strategy provides a prioritized agenda for scientific research and monitoring within the PNMS for the next ten years. The Strategy was developed in parallel with a broader planning effort for the PNMS 2023–2028 strategic plan and in alignment with PNMS core goals: healthy ocean populations, food security and sustainable development. Science approaches outlined in this strategy will provide the information needed to determine the social, economic and ecological status of the PNMS now and under a changing climate, as well as guide management and conservation. This document is intended to delineate the research goals, objectives, activities, and indicators relevant to those with an interest in science, monitoring, management, or outreach relevant to the value of the PNMS.

The strategy is delivered through two primary sections. The first section provides an overview of the document as well as background information on the events and resources that informed the development of its content. Section one concludes with a site description including an overview of existing data resources, ecosystem connections, pelagic fisheries, human dimensions, economic outcomes, cultural/governance outcomes as well as perceptions and expectations.

The second section presents the science and monitoring framework, beginning with an explanation of the engagement process that shaped it — including information about stakeholder engagement and expert working group events that guided the compilation. Section two describes the elements that make up the resulting science and monitoring framework, including the prioritization criteria and research indicators. As the strategy is intended to evolve dynamically while also directly informing active decision making, section two concludes with an adaptive management framework as well as pathways to consider when setting management objectives.



Section 1

Introduction and Background

1.1 PNMS science and monitoring strategy overview

The overall management and broader context of the Palau National Marine Sanctuary (PNMS) is contained within the PNMS 2023-2028 strategic plan (PCS, 2023) which is comprised of overarching and sector-specific strategies which hone in specifically on the enabling environment and coordination while acknowledging existing sector-specific strategies that are being implemented by respective mandated agencies. This science and monitoring strategy is one such sector-specific strategy and was designed to guide research and information generation that is critical for PNMS management now and over time. In doing so it acts as a blueprint detailing scientific research and monitoring priorities for the PNMS between 2024 and 2034.

The aims of the PNMS science and monitoring strategy are to provide the following information:

- Synthesised site description of the PNMS and key scientific references.
- Prioritized information and data needs that are important to advancing the PNMS goals in the next 10 years.
- Knowledge goals, research objectives, and research activities that will provide the data and information needed to achieve the PNMS goals in the next ten years.
- A framework used to prioritise research objectives and activities identified in this strategy.
- Indicators to measure the impact of research objectives and activities identified in this strategy.
- An adaptive management framework that will provide a path for periodic review and revision of this strategy.

The first section of this strategy (Section 1.2) details the background of science and monitoring within the context of the PNMS. This includes an overview of relevant legislation, strategic plans, and expert guidance that informed and helped develop science and monitoring of the PNMS since its inception.

Section 1.3 provides a synthesized site description for the PNMS and includes information on oceanography, biological diversity, pelagic fisheries and human

dimensions. This section aims to provide an initial understanding of current knowledge of the PNMS, including key scientific references.

Section 2.1 describes the science and monitoring strategy development process and details the steps taken to develop the final science and monitoring framework.

Section 2.2 includes the science and monitoring framework for the PNMS, with detailed descriptions of the research objectives and activities. Additional context behind each specific identified objective is also provided.

Prioritization of research objectives and activities was conducted using a prioritisation framework which is described in Section 2.3. Priorities in this strategy were also informed by Palau's reporting commitments under the UN Conventions on Biological Diversity (CBD) and Framework Convention on Climate Change (UNFCCC) as well as regional Micronesia Challenge commitments. In addition, research indicators are provided in this section, to ensure the effective tracking of research objectives and activities.

The proposed strategy review process as part of an adaptive management approach is detailed in Section 2.4.

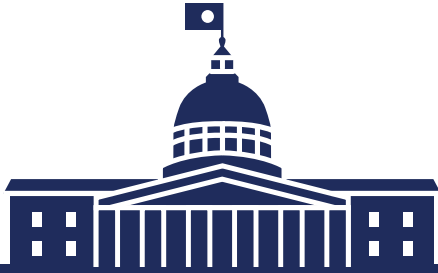
Section 2.5 describes management agencies with a mandate to operate in the PNMS, and details decision pathways through which those agencies may utilize information arising out of this science and monitoring strategy. This section is designed to guide researchers developing partnership research projects with outcomes that are intended to inform management.

Section 3 provides a detailed reference list of all cited resources.

The final section (Section 4) includes various appendices:

1. PNMS science and monitoring expert workshop attendees
2. Research permitting guidance and key contact information
3. Linkages between the PNMS science and monitoring strategy 2024-2034 and the PNMS strategic plan 2023-2028

GOVERNMENT AGENCY



If you are a government agency, this strategy will provide the PNMS science and monitoring priorities for 2024-2034, which can be used to align your agency priorities with, guide future funding proposals and provide an understanding of what science and monitoring will be taking place within the PNMS for the next ten years.

SCIENTIST



If you are a scientist, this strategy will help guide and inform your project development as well as funding applications, to ensure that your research aligns with PNMS priorities. In addition, this strategy will provide a basic understanding of current knowledge of the PNMS, key references and guidance of research processes such as permit applications.

**How will this strategy
be useful to you?**

FUNDING AGENCY



If you are a funding agency, this strategy will enable you to review funding proposals against Palau's research priorities for the PNMS, to ensure that funded research generates needed and useful information according to PNMS research priorities.

COMMUNITY STAKEHOLDER



If you are a community stakeholder, this strategy will give you an understanding of the types of research that will be prioritized within the PNMS for the next ten years.

This strategy is intended to be used as a blueprint for PNMS science and monitoring in the next ten years (2024-2034).

This strategy details Palau's overarching science and monitoring priorities for the PNMS, and not the priorities of any specific entity or organization.

What this strategy IS:

This strategy should make it easy for scientists and collaborators to develop research that aligns with Palau's research priorities for the PNMS.

This strategy is a working document and will undergo timeous reviews to ensure research priorities remain relevant and on track.

This strategy is not intended to limit research not aligned with priorities. Rather, it intends to guide future research development by highlighting what is needed by Palau. The aim is to nudge future research within the PNMS to answer some of the priority PNMS research identified in this strategy.

This strategy is focused on identifying science and monitoring priorities for the PNMS, not funding these priorities. This strategy can be used to help leverage funding support.

What this strategy is NOT:

This strategy is not focused on PNMS management, surveillance, education, outreach or financing.

Regulations, policies, regional agreements and fisheries economics are not included in this strategy. Findings from the research prioritized here can, however, help guide and inform pelagic fisheries development and related regulations, policies and agreements.

1.2 PNMS science and monitoring background

In 2015 the Republic of Palau took a giant step towards large scale marine conservation with the enactment of the Palau National Marine Sanctuary (PNMS) Act and resulting amendment to Palau National Code (PNC) Title 27 (RPPL 9-49, 2015). This act committed to the establishment of a fully protected no-take marine protected area (the Sanctuary), which protects 80% of Palau's Exclusive Economic Zone (EEZ) as well as a Commercial Fishing Zone and Artisanal Fishing Zone within the remaining 20% by 2020. An essential motivation for the development of the PNMS was to unlock the value and benefits of Palau's ocean environments for Palauans, encompassing conservation, food security and sustainable development.

Science formed a foundational building block for the establishment of the Palau National Marine Sanctuary and since its establishment and full implementation, science has continued to guide this process. Science and monitoring were a primary focal area in the first PNMS strategic plan (2017–2021) and a guiding principle for the PNMS was that: “Natural and social sciences should inform the development of actions relating to and within the PNMS.” According to the 2017 PNMS strategic plan (PNMS Office, 2017), the development of a PNMS science and monitoring strategy that will guide future science and monitoring within the PNMS was paramount. This proposed science and monitoring strategy needed, at the very least, to:

- Characterize the biological and ecological features of the PNMS.
- Describe connectivity of biological and ecological systems in the PNMS.
- Describe climate change impacts of the PNMS.
- Describe other anthropogenic impacts (livelihood development, invasive species, fisheries effects, marine debris).
- Fisheries management guided by science and the development of Palauan scientific capacity were also identified as additional priorities.

During the winddown period, from initial demarcation (2015) towards full implementation of the PNMS in January 2020, PNC Title 27 was amended in 2019 (RPPL 10-35, 2019) to allow export of longline and free school purse seine fish caught in the Commercial Fishing Zone. In addition, the amendment included the following organizational responsibilities:

- The Ministry of Agriculture, Fisheries and Environment (MAFE) (previously MNRET) is responsible for the overarching management of natural resources in the PNMS.
- The Ministry of Justice is mandated to protect the safety of persons and wildlife within the Republic's jurisdiction, including restrictions pertaining to the EEZ, the DFZ and the PNMS.
- Palau International Coral Reef Center is assigned the role of administrator for the PNMS and the coordination of scientific research, education and outreach.

In accordance with this mandate, Palau International Coral Reef Center (PICRC), together with Stanford Center for Ocean Solutions (COS), coordinated an expert working group in 2019 and developed a report —at the request of then-President Tommy Remengesau Jr.—with the following objectives (PICRC & COS, 2019):

- Distill and clarify the PNMS legislation.
- Synthesize knowledge about social, ecological, and economic dimensions of the PNMS.
- Conduct new analyses to begin filling knowledge gaps.
- Present informed policy options for achieving desired outcomes.
- Identify priorities for future research, monitoring, and evaluation (resulting in a draft science and monitoring plan).

By distilling the PNMS legislation, PICRC and COS identified three core goals for the PNMS:

1. Healthy Ocean Populations and Ecosystems
2. Food Security
3. Sustainable Development

To operationalize these three core goals,
the PNMS strategic plan 2023-2028 identified six sector goals:

GOAL 1: ENABLING ENVIRONMENT AND COORDINATION

Strengthened capacity and ability of core partner institutions to responsibly steward the PNMS and deliver on their related mandates, in an increasingly improved and supportive enabling environment.

GOAL 2: FOOD SECURITY

Increased food security for the Palauan people, both from increasing production, access, and affordability of PNMS-sourced proteins and from increased economic gains for government, businesses, and individuals.

GOAL 3: ECONOMIC SECURITY

Economic stability and sustainability for the Palau Government, Palauan fishers, and for the Palau-based business and nonprofit sector, both from an economically viable and profitable offshore fishery but also from maintaining strong regional and international positions.

GOAL 4: SAFETY AND SOVEREIGNTY

A safe and secure Exclusive Economic Zone (EEZ) where national and international laws and best practices are respected and followed and negative practices such as Illegal, Unreported and Unregulated (IUU) fishing, bycatch, overfishing, trafficking, and marine debris are minimized.

GOAL 5: SOCIETY AND CULTURE

Maintained and continually growing Palauan pride, knowledge, support, and appreciation for their cultural and natural heritage and its connections to the ocean, such that Palauans take purposeful steps to capitalize on opportunities to achieve benefits from the PNMS and a 100% sustainably managed ocean.

GOAL 6: BIODIVERSITY AND ECOSYSTEMS

Healthy and abundant marine biodiversity and habitats and functioning and resilient ecosystem services are able to support Palauan wellbeing today and, in the future, even in the face of globally changing climates and economies.

This science and monitoring strategy is thus aligned with both the three overarching PNMS core goals as defined in the legislation and the sector goals identified in the 2023-2028 PNMS strategic plan.

Appendix 4.3 provides an overview of linkages between

this strategy and the PNMS strategic plan 2023-2028 (PCS, 2023).

A PICRC & COS draft science and monitoring plan (PICRC & COS, 2019) structured key research questions around the three PNMS core goals and various subgoals:

1. Healthy Ocean Populations and Ecosystems for Palau: Sustaining pelagic marine resources that benefit Palauan livelihoods and drive the Palauan economy

Subgoal 1: Rekindle Palauan societal connection to and appreciation for Palau's offshore environments and resources. Create the next generation of Palauan leadership to manage Palau's open ocean resources.

Subgoal 2: Protect pelagic populations and preserve marine biodiversity in Palau's waters.

Subgoal 3: Support sustainable fisheries by replenishing Palau's fishing zones and adjacent areas.

2. Food Security for Palau: Ensuring sustained, nutritious, and safe supply of food for Palauan residents

Subgoal 1: Increase the availability of and access to domestic pelagic fish according to standard guidelines for health and nutrition for all Palauan residents.

Subgoal 2: Reduce fishing pressure on reef fish for the sustained cultural and socioeconomic benefit to Palauans.

3. Sustainable Development for Palau: Developing a domestic pelagic fishing industry and supporting existing sustainable tourism initiatives

Subgoal 1: Enhance the contribution of the Domestic Fishing Zone to economic development, food security and the conservation of the coral reef fish and invertebrate populations.

Subgoal 2: Support Palau's brand as a high-value ecotourism destination.

Subgoal 3: Support long-term health and well-being for Palauan residents.

Subgoal 4: Support long-term economic sustainability in Palau.

The Palau International Coral Reef Center commenced updated integrated planning for PNMS in 2021, starting with overarching strategic planning in 2021 and science and monitoring planning in 2022. The overarching strategic planning was led by PICRC, the Ministry of Agriculture, Fisheries and the Environment (MAFE) and the Division of Maritime Security and Fish & Wildlife

Protection in Ministry of Justice (MOJ). The science and monitoring planning was led by PICRC with Palauan and research community consultations. Thus, this PNMS science and monitoring strategy is aligned with the PNMS strategic plan 2023-2028 and details the PNMS's science and monitoring priorities for the next 10 years (2024-2034).

Palau has adopted a working definition of "offshore" to indicate any ocean environments that are deeper than the 100-meter-depth contour line out to the international border (Micronesia Challenge, 2006). This definition is in line with regional commitments such as the Micronesia Challenge. In some cases, the 100-meter-depth contour line may be relatively close to shore. The working definition of offshore is different from the legal definition of the PNMS, which goes from the 12-nautical-mile mark to the international border.

1.3 PNMS site description

Overview

The Palau National Marine Sanctuary (PNMS) covers an area of 592 066 km², of which 80% (475 077 km²) is demarcated as a no-take marine protected area (MPA) called the Sanctuary (Figure 1). The Sanctuary, located between 24 NM offshore and Palau's EEZ border, is one of the largest no-take marine protected areas in the world and one of only a handful of large-scale marine protected areas that exclusively include offshore, pelagic and deep-sea environments. In addition to the Sanctuary,

the remaining 20% of the PNMS is zoned as a Domestic Fishing Zone and includes two types of fishing zones: the Artisanal Fishing Zone (12 000 km²), located between 12 and 24 NM offshore, and the Commercial Fishing Zone (105 000 km²), located on the western side of Babeldaob starting from 24 NM offshore to the edge of Palau's EEZ. As per PNMS legislation, commercial longline and free school purse seine fishing is permitted within the Commercial Fishing Zone by both domestic and foreign fleets. In addition, commercial pole and line fishing are allowed in the Artisanal Fishing Zone.

Limited high resolution bathymetric data are available for the entire PNMS, with data from a few cruises publicly available:

Roger Revelle, 2013	https://www.ngdc.noaa.gov/ships/roger_revelle/RR1309_mb.html
Roger Revelle, 2014	https://www.ngdc.noaa.gov/ships/roger_revelle/RR1402_mb.html
Roger Revelle, 2015	https://www.ngdc.noaa.gov/ships/roger_revelle/RR1515_mb.html
Roger Revelle, 2016	https://www.ngdc.noaa.gov/ships/roger_revelle/RR1606_mb.html
Pressure Drop, 2019	https://www.ngdc.noaa.gov/ships/pressure_drop/PD19PT02_mb.html
Sally Ride, 2019	https://www.ngdc.noaa.gov/ships/sally_ride/SR1914_mb.html
Additional bathymetric data can be sourced from the General Bathymetric Chart of the Oceans (GEBCO)	https://www.gebco.net/data_and_products/gridded_bathymetry_data/

Based on available bathymetric data, the Commercial Fishing Zone, located in the Palau Basin, is relatively featureless with large expanses of flat and deep ocean floor. The north and south sections of the PNMS form part of the Kyushu-Palau Ridge and are shallower with various ridges and seamounts. The Palauan trench is located within the eastern part of the PNMS, with the trench axis roughly 19 NM from Palau's eastern shore (Kobayashi et al., 1997; Lemenkova, 2019). The Palauan Trench can be divided into two sections: the north-south trending northern segment with depths decreasing southward from 8000 m to 6800 m and the south-central segment with a relatively constant depth of 6500 m (Kobayashi et al., 1997). Various Japanese-led

research cruises have investigated the Palauan Trench in the past (Kobayashi et al., 1997; Kobayashi, 2004) and recently an expedition led by Victor L. Vescovo on the Pressure Drop conducted the first manned dive to the bottom of the trench in 2022. According to data from the global distribution of seamounts and knolls (Yesson et al., 2011), the PNMS has a total of 211 seamounts ranging in depths from 700 m to 4600 m. Sea knolls are more abundant, with a total of 285 ranging in depths from 1800 m to 5700 m. Even though some bathymetric mapping has been done within the PNMS, large areas still remain unmapped and unexplored. High resolution mapping is also needed to help characterise important features.

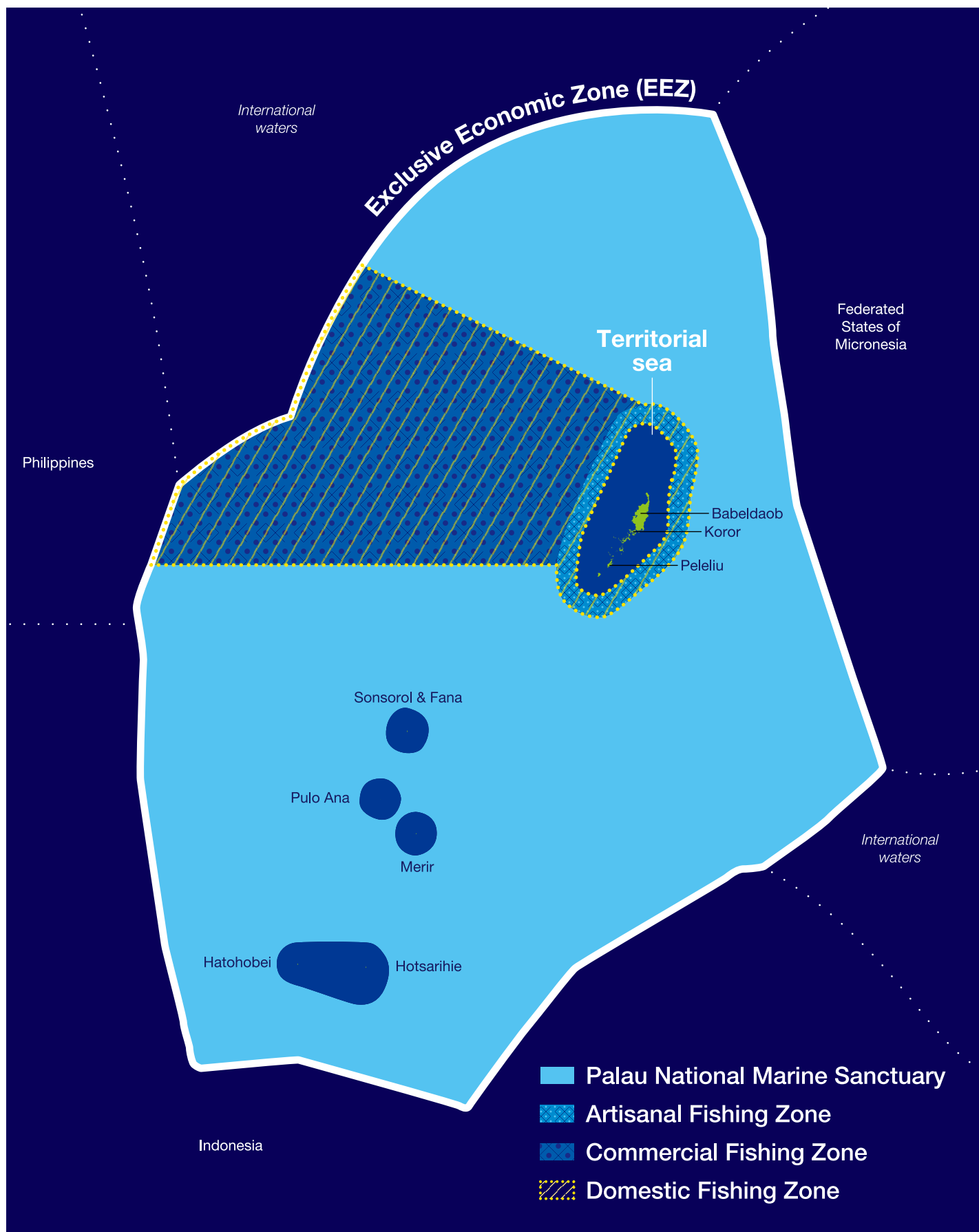
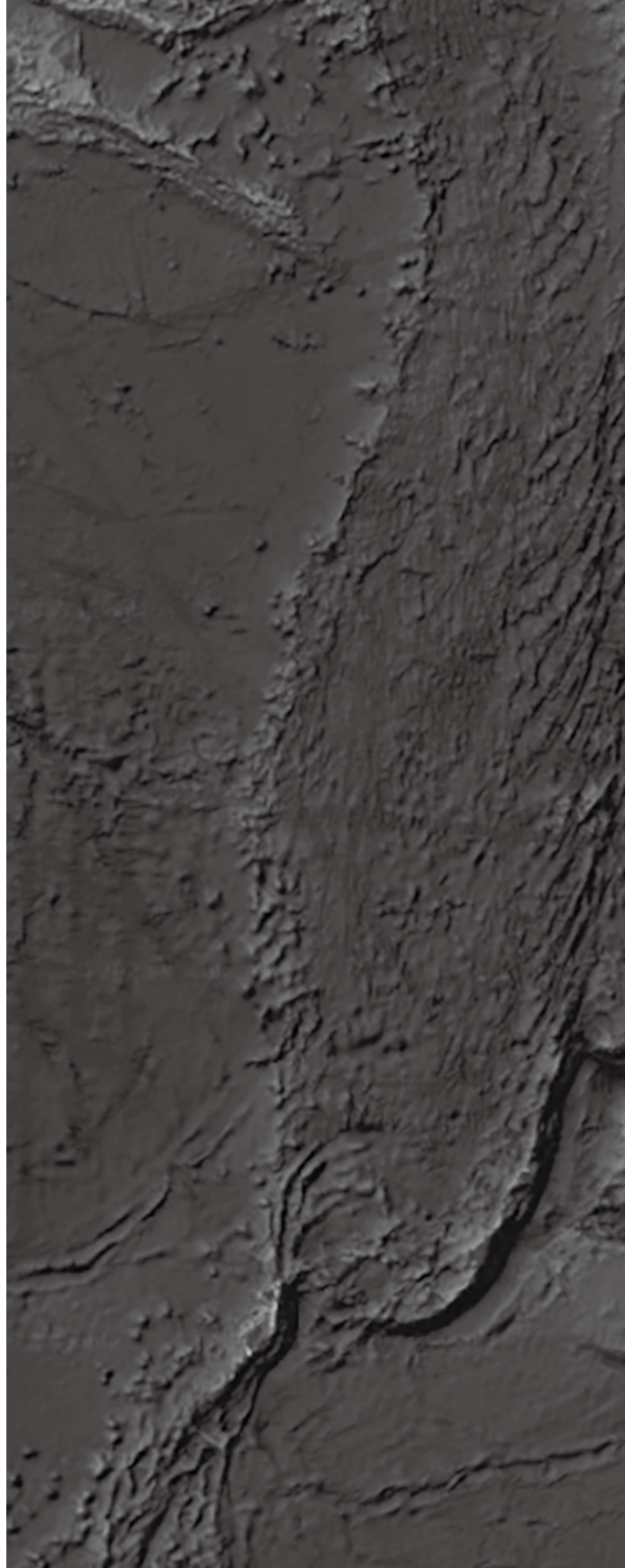


Figure 1: The location of the different management zones within the Palau National Marine Sanctuary: Sanctuary, Commercial Fishing Zone and Artisanal Fishing Zone. (Zoning information provided by PALARIS).

Physical, chemical and biological oceanography

The western tropical Pacific Ocean is characterized by high sea surface temperatures, hence referred to as the Pacific warm pool. Palau, located in the Pacific warm pool, is therefore susceptible to warm water migration. The Kyushu-Palau Ridge lies in the path of the westward-flowing Northern Equatorial Current (NEC) and the eastward-flowing Northern Equatorial Counter Current (NECC) (Johnston et al., 2019). Physical and oceanographic properties within the PNMS generally follow a seasonal pattern, with winter (December-February) characterized by colder sea surface temperatures, a deeper mixed layer depth and lower chlorophyll *a* concentrations (Cimino et al., 2019). Spatial patterns are also present, with the north characterized by lower sea surface temperature and deeper mixed layer depths, and the south with higher chlorophyll *a* concentrations. It is also well known that the Pacific warm pool migrates east to west with El Niño-Southern Oscillation (ENSO) (McPhaden and Picaut, 1990). Thus, these oceanographic patterns are commonly disrupted by ENSO, with deeper mixed layer depths, lower sea surface temperatures and higher chlorophyll *a* concentrations during El Niño periods (Cimino et al., 2019). The western tropical Pacific Ocean is also referred to as a water mass crossroads, because the low latitude current system brings higher-latitude waters. The South Equatorial Current (SEC) and the North Equatorial Countercurrent (NECC) with the Mindanao Eddy (ME) are major components that affect the circumstances within the PMNS (Fine et al., 1994). Currents and ocean circulations surrounding Palau are influenced by the NECC and the ME on a seasonal basis: The ME influences currents to the west of Palau through December to April, curving from northeast to northwest with increasing latitude; as the ME migrate northwards through this period, directions of the currents turn clockwise. From April onwards the ME decreases in its meridional extent reducing its effect on Palau. A cyclonic eddy, named the Palauan Eddy, develops to the northeast of Palau in April where it directs currents along the east coast of Palau to the south with speeds up to 25 cm/s. With the southward migration of



the NECC tail in June, the currents east of Palau are less directionally defined and reduced in magnitude to less than 15 cm/s. On the west of Palau, the ME continues to weaken, with north directed currents of 10-15 cm/s. July to September shows less influence from major surface features and circulation becomes more disordered. Currents during this time are less strong at 5 cm/s. With the regeneration of the Palauan Eddy in October, the south-directed current along the east coast increases to 15 cm/s. In November the currents begin the transition to their north-directed winter pattern with the return of the NECC tail to its northern path (Heron et al., 2006). Understanding how these processes are changing under global climate change is an emerging need.

The US Office of Naval Research Flow Encountering Abrupt Topography (FLEAT) initiative examined how island chains and submerged ridges affect open ocean current systems, from large (hundreds of kilometers) to small (millimeters) scales (Johnston et al., 2019). The NEC varies on synoptic to seasonal to interannual scales, with the ENSO causing the greatest variability (Qiu et al., 2019; Schönau & Rudnick, 2015). During the FLEAT observational period in 2015, an exceptionally large El Niño caused the NECC to strengthen. In 2016, as the El Niño decayed, the NECC turned northward, flowing around Palau (Schonau et al., 2019). The FLEAT program also found an active, wind-driven surface to thermocline layer (roughly the upper 200 m) and a variable sub-thermocline layer during this period. Below the westward-flowing NEC, the North Equatorial undercurrents flow eastward at 9.6°N and 13.1°N, varying considerably in strength and location (Qiu et al., 2019; Schönau & Rudnick, 2015). The dynamics of the undercurrents are less understood; their magnitude is less than and often in the opposite direction of the surface flow. However, at times, the arrival of westward-propagating Rossby waves and eddies disrupt the flow pattern with impacts on the velocity field at depths of up to 4,000 m (Andres et al., 2020).

Biodiversity and ecosystem connections

A dearth of ecological data and information exists for the PNMS, with only a handful of studies focused on the occurrence, ecology and status of species. According to a review of species ranges and reported occurrences within pelagic or deep-sea environments, an estimated 794 vertebrate species can potentially be found within the PNMS, which includes an estimated 565 teleost fishes, 60 sharks and rays, 27 mammals, 20 seabirds and 7 reptiles (PICRC & COS, 2019). In addition, the ranges of nine IUCN Red List-threatened species overlap with the PNMS, including Critically Endangered hawksbill turtles (*Eretmochelys imbricata*) and leatherback turtles (*Dermochelys coriacea*), and Endangered sei whales (*Balaenoptera borealis*), blue whales (*Balaenoptera musculus*), green turtles (*Chelonia mydas*), whale sharks (*Rhincodon typus*), shortfin mako sharks (*Isurus oxyrinchus*), longfin mako sharks (*Isurus paucus*), and great hammerhead sharks (*Sphyrna mokarran*). A marine mammal survey conducted in Palau in 2012 (Andrews, 2013) confirmed the presence of eight mammal species within Palau's waters: pygmy killer whales (*Feresa attenuata*), melon headed whales (*Peponocephala electra*), sperm whales (*Physeter macrocephalus*), spinner dolphins (*Stenella longirostris*), pantropical spotted dolphins (*Stenella attenuata*), false killer whales (*Pseudorca crassidens*), short finned pilot whales (*Globicephala macrorhynchus*) and beaked whales (*Unidentified Ziphius/Mesopledon*). Anecdotally, an additional four mammal species can be found within Palau: risso's dolphin (*Grampus griseus*), killer whale (*Orcinus orca*), bryde's whale (*Balaenoptera edeni*) and striped dolphin (*Stenella coeruleoalba*). These species most likely reside within or migrate through the PNMS during some period of their lives. For example, two whale sharks tagged with satellite tags in Cenderawasih Bay, Indonesia, were found to spend time within Palau's EEZ, before returning home (Meyers et al., 2020). A recent review by Hari et al. (2021) listed 28 species of sharks and rays that occur or can potentially occur within Palau's pelagic and deep-sea environments. Both green turtles and hawksbill turtles have been found to travel long

distances, many returning to their breeding site (Helen Island) (Klain et al., 2007). Preliminary findings from a 2022 mid-water baited remote underwater video (BRUV) survey within the PNMS recorded a turtle at least 30 NM north of the main island of Palau.

The only deep-sea research conducted within Palau was restricted to areas that fall outside the PNMS (<12 NM from Palau's main island) (Friedlander et al., 2014; Claassens et al., Under Review). Findings from this research provide an initial understanding of deep-sea species that occur within Palau, and most likely extend into the PNMS. Friedlander et al. (2014) deployed 24 deep-sea cameras in depths ranging from 260 to 3500 m and recorded 26 fish taxa from 19 families. Cutthroat eels from the family Synaphobranchidae were most common, followed by rattails from the family Macrouridae (Friedlander et al., 2014). A full list of all fishes and invertebrates recorded during this expedition is available in Friedlander et al. (2014). During a subsequent deployment of two deep-sea cameras in 2021 (1288 m and 643 m respectively) (Claassens et al., Under Review) a sleeper shark *Somniosus* sp. was observed on the east coast of Palau. This was both a first record of *Somniosus* sp. for Palau as well as a range extension for this genus.

Various economically important fish species occur in the PNMS, bigeye tuna (*Thunnus obesus*) and yellowfin tuna (*Thunnus albacares*) being the most important. Prior to the establishment of the PNMS, Japanese (distant water fleet) and Taiwanese (locally based fleet) fleets dominated the tuna fisheries within Palau. The location of fishing effort for each flag state were found to vary seasonally as well as inter-annually based on the El Niño–Southern Oscillation (ENSO) (Cimino et al., 2019). Fishing effort did, however, seem to remain high throughout the year off the southeast of Palau's main island (particularly for the Taiwanese fleet), suggesting that this zone consistently contains high fish abundances (Cimino et al., 2019; PICRC & COS, 2019). Bigeye tuna consistently dominated catches in the North, while yellowfin tuna dominated Southern catches (Cimino et al., 2019). Through the development of an integrated habitat index for *T. obesus*, Li et al. (2012) concluded that most adult fish can be found at depths of less than 200 m, and that the effect of environmental variables on fish distribution vary across depths. Based on these models, temperature and dissolved oxygen were found to influence the distribution of *T. obesus* most (Li et al., 2012). Cimino et al. (2019) also found that various ocean features and processes influence tuna fishing locations:





core fishing regions did not overlap with the center of the North Equatorial Counter Current (NECC); higher absolute dynamic topography (0.8-1.2 m) (which acts as an indicator of the thermocline within Palau) was preferred, and year-round deep mixed layer depths were avoided. Gilman et al. (2016) investigated the effect of various variables on the standardized catch per unit effort for both *T. obesus* and *T. albacares*. For *T. obesus* he found that the number of hooks per set (more hooks equaled higher catch); time (a decreasing trend in CPUE between 1999 and 2011) as well as time of day (peak catches at 3 pm, lowest catches at 2 am) were significant variables to determine catch rates. For *T. albacares*, the number of hooks per set (more hooks equaled higher catch); time of day (peak catches at 4:30 am, lowest catches at 5:30 pm) and lunation (highest catch rates occurring around day 3 of the synodic lunar month, lowest catch rates occurring at around day 21 of the synodic lunar month) were found to be significant variables to determine catch rates (Gilman et al., 2016).

Surveys conducted in the North Pacific between 1992 and 1996 recorded high abundances of juvenile *Thunnus* spp. in the area of the North Equatorial Counter Current, while juvenile skipjack tuna *K. palamis* were widely distributed from the North Equatorial Current to the North Equatorial Counter Current (Tanabe et al., 2001). The authors concluded that the offshore environments of Palau and Micronesia are important for larval and juvenile tunas as both a nursery area and at early life stages (Tanabe et al., 2001). A recent study tracking large pelagic predators within the PNMS using satellite tags found that *T. albacares* travel extensive distances, up to 2010 km, within the region and commonly travel outside the PNMS (Filous et al., 2022). Blue marlin (*Makaira mazara*) tagged within Palau's Artisanal Fishing Zone were found to have more restricted dispersal patterns and returned to the PNMS (Filous et al., 2022). Additionally, juvenile *T. albacares* were found to commonly utilize nearshore fish aggregation devices (FADs) around Palau, especially deeper FADs (Filous et al., 2020). Numerous questions remain about the status, distribution, projections, and sustainability of fish in the PNMS.



Pelagic fisheries

All commercially targeted tuna species in Palau are considered to be part of a western and central Pacific Ocean regional stock and are thus managed on a regional scale. The most recent stock assessments for tuna conducted by the Commission for the Conservation and Management of Highly Migratory Fish Stocks in the Western and Central Pacific Ocean (WCPFC) took place in 2019 and 2020. According to these assessments, both *T. obesus* (bigeye tuna)

(<https://www.wcpfc.int/doc/01/bigeye-tuna>)

and *T. albacares* (yellowfin tuna) (<https://www.wcpfc.int/file/588254/download?token=jcp-yYHK>) stocks (assessed in 2020) are not currently overfished regionally. The fishing impact for both species was, however, found to be higher in tropical regions (which includes region 7 within which Palau falls), particularly for juvenile *T. obesus*. According to the North Pacific Albacore assessment, albacore tuna (<https://www.wcpfc.int/doc/05/north-pacific-albacore-tuna>) (*Thunnus alalunga* assessed in 2020) is likely not overfished relative to the limit reference point adopted by the Western and Central Pacific Fisheries Commission.

Katsuwonus pelamis (skipjack tuna) (<https://meetings.wcpfc.int/file/11266/download>) (assessed in 2022) was found to be moderately exploited in the region, with an estimation that the stock is becoming increasingly depleted over time, a trend which is largely driven by the equatorial region. Both swordfish *Xiphias gladius* (<https://www.wcpfc.int/doc/07/north-pacific-swordfish>) (assessed in 2018) and *Makaira nigricans* (blue marlin) (<https://www.wcpfc.int/doc/11/pacific-blue-marlin>) (assessed in 2021) show no evidence of excess fishing mortality above the fishing pressure corresponding with maximum sustainable yield (Fmsy) or depletion of spawning potential. Catch data from Palau collected from 1999 to 2011 found that yellowfin and bigeye tuna made up 26% and 12% of the total catch, respectively, and were the top two species of both the total and retained catch (Gilman et al., 2015).

All licensed commercial fisheries catch needs to be logged and reported to the Government of Palau and The Pacific Community (<https://www.spc.int/>), although mis- and underreporting is commonplace (Brown et al., 2021). Electronic monitoring is a relatively new attempt at validating conventional catch log data, and

an electronic monitoring trial on tuna longliners was conducted in Palau, Federated States of Micronesia and the Republic of the Marshall Islands between 2016 and 2018 (Brown et al., 2021). Catch rates were generally under-reported when using logbooks, with yellowfin tuna catch rates found to be 30% higher when using electronic monitoring. Species diversity was also found to be higher when using electronic monitoring (Brown et al., 2021). In Palau, the shark discard rate recorded by using electronic monitoring was 7.7 times higher than recorded by logbooks. In addition, the same study found that turtle discard was higher when recorded by electronic monitoring compared to logbooks (Brown et al., 2021).

Commercial shark fishing has been banned in Palauan waters since 2003 (Gilman et al., 2015; [27 PNCA 127](#)) and all shark and ray bycatch from commercial fishers is required to be logged for reporting purposes and then released or discarded. According to Gilman et al. (2015), sharks made up 17% of the total catch between 1999 and 2011, of which 98% were discarded (Gilman et al., 2015). Reported longline bycatch between 2010 and 2020 resulted in the catch of 2,122,279 individuals from 117 taxa, dominated by pelagic stingrays (18,890 individuals) (Jaiteh et al., 2021). A target-catch:bycatch ratio of 1:1 was found and bycatch rates, retention rates and species composition varied between different fleets (Japanese vs. Taiwanese fleet) (Jaiteh et al., 2021). In addition, post-release mortality rates for blue sharks (*Prionace glauca*) and silky sharks (*Carcharchinus falciformes*) were found to be 0.17 and 0.20 respectively, with 87% of mortalities occurring within 2 days of release (Musyl & Gilman, 2018).

After the establishment of the PNMS in 2015 and the subsequent winddown period between 2015 and 2020, longline tuna fishing rates within the proposed Commercial Fishing Zone increased from 7% to an estimated high of 18% (PICRC & COS, 2019). After the enactment of the PNMS in 2020 and the departure of the Taiwanese fleet, commercial free-school purse seine and longline tuna fishing permitted within the Commercial Fishing Zone was dominated by Japan and Palau.





PNMS human dimensions

The human dimensions research surrounding the PNMS to date has focused on understanding its current, potential, and expected social outcomes; perceptions of the PNMS; and key stakeholders and the reasons for their support and/or opposition. Additional understanding of socioeconomic and cultural dimensions is needed.

Economic outcomes

Economic impacts related to fisheries are a key expected outcome from the PNMS and are a priority for the Government of Palau. To this end, socioeconomic research related to the PNMS has focused on fisheries and related economic impacts from the establishment of the PNMS (Dacks et al., 2020, Lewis et al., 2020). A fishery value chain survey conducted between 2016 and 2018 concluded that most reef fish are consumed by local residents, while most tuna caught by the locally based longline fishery are exported (Dacks et al., 2020). Most tourists consumed pelagic fish (PICRC & COS, 2019; Lewis et al. 2020). Additionally, Lewis et al. (2020) concluded that the PNMS might increase pelagic fish prices, which could increase reef fish consumption by tourists. Offering a sustainable brand of pelagic fish to tourists may be useful to help curtail this potential negative impact, as tourists were willing to pay more for a sustainable tuna brand (Lewis et al., 2020). Ongoing research is, however, needed to determine if the above-mentioned predicted impacts on the fishery supply, demand and economics have in fact been realized.

Palau is part of the Parties of the Nauru Agreement (PNA) and as such subscribes to PNA's Vessel Day Scheme (VDS). Gruby et al. (2020) investigated interactions between the PNMS and the VDS. They explain how the VDS influenced the original design of the PNMS to ensure that Palau could maintain its VDS allocations, and they also consider how Palau's participation in the VDS helps offset potential losses in fishing revenue related to the PNMS. Finally, they consider how the VDS could affect the PNMS's outcomes for tuna conservation, as it relates to the geography of fishing displacement and the



spatial ecology of tuna at national and regional scales. A study by Villaseñor-Derbez et al. (2020) similarly found that trading of vessel days within the PNA allows large-scale conservation action, such as the PNMS, to take place without excessive cost to the conserving country. In addition, the authors concluded that the conventional way of determining fishing rights under the PNA (based on a formula that includes past fishing effort and biomass of the participating country) will result in high long-term economic losses for the conserving country (Villaseñor-Derbez et al., 2020). A formula using only biomass will, however, lower these costs to under 1%, essentially eliminating the cost of conservation (Villaseñor-Derbez et al., 2020). Continued examination of socioeconomic consequences of the PNMS is needed.

Cultural/governance outcomes

The PNMS has also led to some important cultural outcomes related to how the Council of Chiefs – the highest-ranking chiefs in Palau – protected the sanctuary through *bul*. Interviewees reported that this was the first time ever that a *bul* was used in a pelagic ocean space. This gave the PNMS a culturally important seal

of approval, and it also formally embedded pelagic conservation within the traditional governance system. This may have future implications for public engagement with the PNMS, as well as who controls pelagic ocean spaces in Palau and through what governance mechanisms (Fairbanks & Gruby, unpublished data).

Perceptions and expectations

A pilot survey to understand community perceptions of the PNMS and the broader suite of benefits it may provide was conducted in Airai in 2020 (Koshiba et al., 2022). A PNMS household questionnaire was developed and structured to focus on: 1) Socio-demographics, 2) Sustainable livelihood and food security, 3) Perceived benefits from the PNMS and 4) PNMS knowledge and support. To test the efficacy of the PNMS household survey, a pilot survey of 266 households was conducted in Airai in 2020. Similar to previous studies (e.g., Dacks et al., 2020), people reported preferring reef fish to pelagic fish, with a greater dependence on cheaper and consistently available reef fish. Most respondents agreed with the potential benefits from the PNMS identified during this study, which includes species conservation,



replenishment of fish resources, benefits to the nearshore environment and maintenance of Palau's conservation culture locally and internationally. Findings also suggest that even though there is high support for the PNMS, communities have a limited understanding of this large-scale marine protected area (LSMPA), which emphasises the need for the development of more effective outreach and education initiatives. Long-term monitoring of potential benefits and perceptions of the PNMS by the people of Palau can now be developed using the standard approach developed here. This will enable tracking of PNMS outreach and education effectiveness, the realisation of potential PNMS benefits to the people of Palau as well as any changes in fish resource use over time (Koshiba et al., 2022). This study is being expanded in 2023 to include all states within Palau.

Gruby et al. (2017) draws on case studies of five proposed and designated LSMPAs globally, including the PNMS, to develop a framework for conceptualizing the unique social outcomes of LSMPAs. With respect to the PNMS, the paper highlights how the PNMS set in motion broad-scale economic and geographic changes through which Palau is reorganizing and reclaiming its EEZ in

an effort to benefit its people at a national scale. The paper also describes specific expected outcomes, such as the expectation among local fishermen that pelagic stocks will expand and come closer to shore in the DFZ, where they can fish without foreign competition. Overall, the paper draws attention to “the need for broader conceptualizations of social outcomes of different kinds, at different scales, and/or for different populations in addition to those most familiar based on work in conventional MPAs” (Gruby et al., 2017).

Finally, Fairbanks and Gruby (unpublished data) presents results from an in-depth case study of the human dimensions of the Palau National Marine Sanctuary conducted between 2015-2018. The report describes the different stakeholder groups that are active in Palau's offshore waters or have distinct interests in the waters, its resources, and its conservation; the early political and institutional history of the PNMS, including why people supported and opposed it; and, finally, the expected and realized outcomes at the time of publication, as well as key outcomes to track over time.

Section 2

Science and Monitoring



2.1 PNMS science and monitoring strategy development

Development of the PNMS science and monitoring strategy commenced in 2022 and was informed by several resources. The 2019 PICRC and COS-led working group report on the PNMS (Managing Ocean Change and Supporting Food Security) provided foundational research and an initial draft for a “Science, Monitoring, and Evaluation Plan” for the PNMS. In 2022, PICRC conducted a detailed literature review of science and monitoring priorities identified in previous PNMS

management plans and reports, past and ongoing PNMS research (both peer-reviewed and gray literature), as well as global large-scale MPA science and monitoring agendas and priorities.

Based on the initial literature review of PNMS science and monitoring as well as global large-scale MPA science and monitoring priorities, five overarching research questions were developed to guide strategy development:

1 What is out there - what important, unique and threatened species, habitats and features are found in the PNMS?

To understand the effectiveness of an MPA, a clear understanding of the baseline status is required. This can include abundance and distribution of organisms, habitats and ecosystems. A baseline understanding is used to measure change over time and the effectiveness of an MPA in achieving goals.

Considering this statement, what is the baseline status of the PNMS?

Marine protected areas can provide much needed adaptation to climate change impacts and changes. Large-scale MPAs have the potential to encompass range shifts of marine species under climate change, reduce cumulative stressors on ecosystems, enabling faster recovery from climatic impacts, promote larger populations more resilient to extinction and with greater genetic diversity, and act as wildlife refugia.

Considering this statement, is the PNMS an effective climate change adaptation strategy and if so, how?

2 What ecological, social and economic climate change adaptations do the PNMS provide locally, regionally and globally?

3 What role does the PNMS play to ensure future food security locally and regionally?

Marine protected areas are a well-known fisheries management approach for food security. Food security was also identified as one of the overarching aims of the PNMS. Tuna and other highly mobile fish are of particular interest and research is needed to determine how the PNMS contributes towards food security for Palau and the region. Focus should also be placed on the future of pelagic fisheries and how the PNMS might play a role in supporting more resilient fish populations.

Considering this statement, is the PNMS an effective strategy for future food security and if so, how?

One of the aims of long-term monitoring is to detect system change over time. What to monitor, how and how often are key questions in the development of an effective long-term monitoring program. The size and remoteness of a system increases the difficulty of effective monitoring – logistically, financially and in terms of the increased complexity of the system and understanding changes over time. The PNMS provides a unique learning opportunity to develop a long-term monitoring program for a large-scale, remote MPA. What should be monitored, what approaches are effective in detecting change, what can be done on a budget etc.

Considering this statement, what long-term monitoring approaches can be developed for the PNMS?

4 What does effective long-term ecological monitoring of a pelagic large scale MPA look like?

5 What are the local, regional and global social, cultural, economic and political implications of the PNMS?

Large-scale MPAs not only have an effect on the environment, but also on social, cultural, economic and political aspects. Direct and indirect implications of large-scale MPAs need to be understood to determine if the MPA is achieving its goals.

Considering this statement, how has/is the PNMS affecting Palau economically, socially, culturally and politically?

The resulting collection of existing information and overarching research questions informed engagement with key local stakeholders as well as a workshop with large-scale MPA experts from around the globe. Workshops with key local stakeholders were held in August and November 2022 aimed at sharing the PNMS science and monitoring strategy development process and providing an opportunity for input and knowledge sharing with local stakeholders. Workshops were structured around the overarching research questions detailed above, which provided a starting point for discussions. Stakeholder discussions and feedback aligned with the proposed overarching research questions and reinforced the approach taken for PNMS science and monitoring strategy development. Information generated during these workshops were used to further refine the overarching research questions and guide engagement with expert stakeholders to ensure local significance and representation.

To ensure that the PNMS science and monitoring strategy was developed with a high level of scientific integrity and rigor, a workshop with large-scale MPA experts from across the globe was held in December 2022 (see Appendix 4.1 for full list of participants).

The aims of this workshop were to:

- Review and refine Palau National Marine Sanctuary science and monitoring priorities (overarching research questions).
- Review and refine Palau National Marine Sanctuary science and monitoring strategy.
- Develop key scientific partnerships to enhance and develop Palau National Marine Sanctuary science and monitoring (knowledge and mentorship hub).
- Identify collaborations and funding opportunities to develop further.

Workshop discussions were guided by the five overarching research questions detailed above. An additional temporal layer was included whereby science and monitoring discussions were framed within three different timeframes: Short term (1-3 years); Medium term (4-6 years); Long term (7- 10 years).

Based on the detailed literature review and feedback and discussions from the local stakeholder and expert engagements, three knowledge goals, eight research objectives and 23 research activities were identified.

Short term research (2024-2026) Research that will generate critical data to inform time sensitive PNMS management needs. In addition, it is logistically viable to conduct this type of research and generate robust scientific results within a short timeframe.	Medium term research (2027-2029) Research that will generate critical data to inform PNMS management needs. Logistically and practically, this type of research and the generation of scientifically robust results requires a longer timeframe.	Long term research (2030-2034) Research that will generate useful data to inform PNMS management needs. This type of research requires a long time-frame to implement.
--	--	--

2.2 PNMS science and monitoring framework

This section details the three knowledge goals, eight research objectives and 23 research activities identified for the PNMS. Context is provided for each research objective. Each research activity has undergone prioritisation (according to the prioritisation framework described in section 2.3), resulting in the advised time frame for each activity included below. Indicators to track the process of research objectives and activities have been developed according to the process detailed in Section 2.3 and have been included below.

Knowledge goal

A knowledge goal refers to the overarching knowledge destination that Palau wants to achieve through science and monitoring work. Each knowledge goal is directly linked to the original core goals of the PNMS: healthy ocean populations, food security and sustainable development.

Research objective

To achieve these knowledge goals, eight research objectives have been identified, which provides the overarching research direction for the next ten years.

Research activity

A research objective has a set of specific research activities which outlines the specific research that should be undertaken to achieve the specific objective. A research activity will be made up of various research projects specifically targeted at generating data to inform the research activity and achieve the research objective.

Research objective and Activity indicator

Each research objective will have an objective indicator to guide and track the implementation and achievement of objectives. Similarly, each research activity will have an activity indicator to guide and track activity implementation.

Activity products

In addition, proposed activity products that will be developed by each activity are listed.

Milestone questions

Finally, each activity has a set of milestone questions that can be asked during strategy reviews, to determine if implementation is on track.

Knowledge goal 1

Healthy ocean populations and ecosystems

In the next 10 years, exploration and scientific research will illuminate and bolster Palau's knowledge to enable sustaining diverse populations and habitats in changing ocean environments

Research objective 1.1

Explore the spatial and temporal occurrence, extent and status of key physical and biological species, habitats, ecosystems, features and processes in the PNMS.

Objective indicator 1.1

By 2034, key physical and biological species, habitats, ecosystems, features and processes are identified, classified and mapped, with key variables and approaches for long-term monitoring established.

Context: Scientific exploration and characterisation is a critical first step for PNMS science and monitoring, owing to the dearth of available scientific information and data for this large-scale MPA. This first objective hones in on the need to map, characterise and understand both the physical and biological realms of the PNMS across space and time. This will not only focus on mapping these realms, but also on generating data to better understand the different physical and biological processes and how they vary on spatial and temporal scales.

Activity 1.1.1

Critically assess and synthesize available data, conduct a gap analysis and expand on analyses existing of data on the PNMS

1-3
years

Activity indicator 1.1.1

By 2026, existing data relevant to the PNMS have been collated, reviewed and synthesised and undergone a knowledge gap analysis to identify where further analyses are required.

Activity products 1.1.1

1. Database with all collated data/studies/research.
2. Knowledge gap analysis report.
3. Research proposals outlining where further analyses can be done with existing data.

Milestone questions 1.1.1

Do we have an overview of research applicable to the PNMS?
Do we know what is missing?
Do we know what needs to be done next?

Activity 1.1.2

Identify and characterize hotspots of productivity and diversity within the PNMS and determine key variables to track over time



Activity indicator 1.1.2

By 2029, hotspots of both productivity and diversity have been identified, characterised and mapped within at least 50% of the PNMS and key variables and approaches for long-term monitoring have been established.

Activity products 1.1.2

1. Spatial and meta-data sets of productivity and biodiversity hotspots.
2. Identification of key variables to track over time, with a monitoring approach developed and implemented.
3. Published articles/reports characterising hotspots of productivity and diversity within the PNMS.

Milestone questions 1.1.2

Do we know where hotspots for productivity and biodiversity are?
Do we have maps of these hotspots?
Do we have enough information to characterise and understand these hotspots?
Can we track changes of these hotspots over time?

Activity 1.1.3

Conduct further exploratory research of identified hotspots



Activity indicator 1.1.3

By 2034, targeted opportunities for exploratory research of identified productivity and diversity hotspots within the PNMS have been identified and supported.

Activity products 1.1.3

1. Collaborative research agreements with external exploration programs and initiatives.
2. Spatial and metadata of hotspots with higher resolution.

Milestone questions 1.1.3

Have we taken advantage of proposed research cruises?
Were we able to guide and inform cruise planning - to achieve our objectives?
Do we know more about the PNMS now?

Research objective 1.2

Investigate key climate change dynamics and impacts within the PNMS and prioritize monitoring activities related to biological, economic and physical conditions.

Objective indicator 1.2

By 2034, the level of understanding of climate change within the PNMS has increased to effectively enable the Government of Palau to plan and adapt to impacts, with key variables and approaches for long-term monitoring developed.

Context: The PNMS is not static, and is constantly changing, especially given global climate change. Research objective 1.2 focuses on understanding the changing nature of the PNMS — specifically the influences, impacts, opportunities, and threats of climate change within the PNMS — on a locally significant level and what the PNMS might look like in the future. This objective will aim to generate information for management to plan and prepare for a future PNMS (on a biological, physical and economic level) under climate change.

Activity 1.2.1

Downscale global climate change models for the Palauan context

1-3
years

Activity indicator 1.2.1

By 2026, the most relevant global climate change models, which include locally significant variables, have been applied to the PNMS.

Activity products 1.2.1

1. Locally significant climate change model outputs for Palau.
2. Understanding what the future state of the PNMS will be under climate change - specifically for Palau/Region.
3. Management guidance.

Milestone questions 1.2.1

Were we able to downscale global models to make it more applicable to Palau and the region?
Do we have a robust projection of what the future PNMS will look like under climate change - economics, ecology, social?

Activity 1.2.2

Document and measure climate change impacts in the PNMS to help prioritize adaptive capacities

4-6
years

Activity indicator 1.2.2

By 2029, major climate change impacts within the PNMS are identified, characterised and quantified, and key variables and approaches for long-term monitoring have been established.

Activity products 1.2.2

1. Identification and quantification of climate change impacts on the PNMS (social, ecological, economic).
2. Determination of planning horizons and scenarios of interest.
3. Key variables and approaches to track this over time.

Milestone questions 1.2.2

Do we know what the impact of climate change will be on the PNMS?
Do we know which impacts are the most critical?
Can we track these impacts over time?

Activity 1.2.3

Identify, characterise and quantify the climate-change-induced losses and damages in the PNMS

4-6
years

Activity indicator 1.2.3

By 2029, Palau has identified scientifically defensible losses and damages that meet the criteria of the UNFCCC; and is tracking additional losses and damages where relevant.

Activity products 1.2.3

1. Quantified list of all climate change induced losses and damages for the PNMS.
2. Key variables and approaches to track this over time.

Milestone questions 1.2.3

What damages and losses are caused by climate change?
Can we track this over time to see if management/policy is effective?

Activity 1.2.4

For identified climate change impacts that become important, conduct forward-looking research to develop adaptive capacity in the face of different climate projection scenarios

7-10
years

Activity indicator 1.2.4

By 2034, forward-looking research for the three most critical climate change impacts identified for the PNMS have been developed and is being implemented resulting in effective, locally significant adaptive measures.

Activity products 1.2.4

1. Practical steps/actions that can be taken by Palau to adapt to potential future climate change impacts.
2. Various project proposals aimed at further understanding the most critical climate change impacts within the PNMS.

Milestone questions 1.2.4

Do we know what the most critical impacts of climate change on the PNMS are?
Have we developed focused forward-looking research to better understand and predict these impacts?

Research objective 1.3

Determine the conservation value of the PNMS to threatened and culturally important pelagic and deep-sea species.

Objective indicator 1.3

By 2034, the conservation value of the PNMS to key threatened and culturally important pelagic and deep-sea species have been identified, characterised and quantified.

Context: In an era of global biodiversity loss, marine protected areas and their proven ability to conserve biodiversity is becoming increasingly important. The location of the PNMS and its sheer size can result in significant biodiversity and species conservation outcomes. Research objective 1.3 will hone in on the role of the PNMS in conserving biodiversity, specifically threatened and culturally important species.

Activity 1.3.1

Identify and quantify the conservation value of the PNMS for key pelagic and deep-sea species

7-10
years

Activity indicator 1.3.1

By 2034, the most important threatened and culturally important species within the PNMS have been identified, and the conservation value of the PNMS to these species have been described and quantified, with key variables and approaches for long-term monitoring developed.

Activity products 1.3.1

1. List of the most important threatened and culturally important species related to the PNMS.
2. The identification and quantification of the conservation value of the PNMS to these species.
3. Key variables and approaches to track this over time.

Milestone questions 1.3.1

Have the most important threatened and culturally important species been identified (list and reasoning)?

Have the different ways in which the PNMS helps to conserve these species been identified?

Can we monitor these species and the role of the PNMS in their conservation over time?

Activity 1.3.2

Explore the potential of the PNMS to conserve global, regional, and local biodiversity



Activity indicator 1.3.2

By 2034, the role of the PNMS to conserve global, regional and local biodiversity has been described and quantified, with key variables and approaches for long-term monitoring developed.

Activity products 1.3.2

1. Suitable metric for biodiversity.
2. Identification of different diversity conservation outcomes of the PNMS.
3. Biodiversity conservation role placed in a regional and global context.
4. Key variables and approaches to track this over time.

Milestone questions 1.3.2

Has a suitable metric for biodiversity been determined?
Have different biodiversity roles of the PNMS been identified?
Can we determine if the PNMS is working as a conservation tool?
Can we track this over time?

Knowledge goal 2

Food Security

In the next 10 years, Palau will have the knowledge it needs to sustainably manage and support economically and culturally important pelagic fish populations as well as pelagic fisheries to benefit Palauans

Research objective 2.1

Understand the dynamics and economics of domestic and regional pelagic fisheries throughout the supply chain.

Objective indicator 2.1

By 2034, available data and information of the dynamics and economics of domestic and regional pelagic fisheries can be effectively used for locally significant, sustainable pelagic fisheries management.

Context: There is a growing trend in promoting the use of pelagic resources within Palau, particularly owing to increased pressure on reef resources. An increased utilization of pelagic resources can potentially play a major part in future food security. Research objective 2.1 will focus on all aspects of pelagic fisheries, untangling the intricacies of supply, demand and the related economic and social implications.

Activity 2.1.1

Investigate the pelagic fishery supply and demand dynamics related to the PNMS

1-3
years

Activity indicator 2.1.1

By 2026, the dynamics and drivers of supply and demand of pelagic fish resources within Palau are understood and useful for effective fisheries management, with key variables and approaches for long-term monitoring developed.

Activity products 2.1.1

1. Data of pelagic fish supply and demand within Palau - ability to predict future scenarios.
2. Ability to run models and scenarios to predict possible outcomes of changing harvest or conservation strategies or changing system inputs or outputs.
3. Key variables and approaches to track this over time.

Milestone questions 2.1.1

Do we have a precise estimate of current supply and demand rates of pelagic fish for Palau?
Do we know what drives these estimates?
Do we have a precise estimate of future supply and demand rates of pelagic fish for Palau?
Have key variables for long term monitoring been identified?
Approaches to monitor these?

Activity 2.1.2

Determine the viability of the Commercial Fishing Zone and Artisanal Fishing Zone to support local food security

1-3
years

Activity indicator 2.1.2

By 2026, data are available to determine the viability of the Commercial Fishing Zone and Artisanal Fishing Zone to support local food security.

Activity products 2.1.2

1. Data showing the potential economic biomass that can be produced in the fishing zones.
2. Data showing the potential fishing effort that can be supported by the fishing zones.
3. An ability to predict future scenarios under different management options.

Milestone questions 2.1.2

- Do we know how much fish the fishing zones can produce?
- Do we know if it is economically viable to only fish in these zones?
- Do we know how this will change in the future?

Research objective 2.2

Determine and track the status and population dynamics of economically and culturally important pelagic fish species.

Objective indicator 2.2

By 2034, enough information is available to determine the status of economically and culturally important pelagic fish populations, with the provision of locally significant management and regulatory guidance where needed.

Context: Marine Protected Areas are a tried and tested fisheries management approach to help protect fish populations to ensure sustainable use. In addition, spill over from protected areas has the potential to support fisheries in adjacent areas. Research objective 2.2 aims to determine and better understand the potential of the PNMS to benefit both pelagic and coastal-pelagic fish populations.

Activity 2.2.1

4-6
years

Determine the spatial and temporal status of pelagic fish populations

Activity indicator 2.2.1

By 2029, the status of key (at least 5 species) economically and culturally important pelagic fish populations are known, locally significant management and regulatory guidance for stocks are developed and key long-term monitoring variables and approaches have been developed.

Activity products 2.2.1

1. Stock status of at least 5 species of economically and culturally important pelagic fish populations.
2. Key variables and approaches to track this over time.
3. Key locally significant management and regulatory advice to ensure sustainable stocks.

Milestone questions 2.2.1

Do we know the stock status of at least 5 pelagic fish species?
Do we know what drives the status of these stocks?
Can we conduct long-term monitoring to track the status of stocks over time?

Activity 2.2.2

1-3
years

Investigate the benefits and role of the PNMS on coastal-pelagic and nearshore fish populations

Activity indicator 2.2.2

By 2026, the role of the PNMS on key (at least 5 species) coastal-pelagic and nearshore fish populations are characterised and quantified, with key variables and approaches for long-term monitoring developed.

Activity products 2.2.2

1. Data on space and resource use of coastal-pelagic and nearshore species in the PNMS.
2. Quantified information of the role of the PNMS for key (at least 5 species) coastal-pelagic and nearshore populations, including baitfish.
3. Key variables and approaches for long-term monitoring developed.

Milestone questions 2.2.2

How is the PNMS used by key (at least 5 species) coastal-pelagic and nearshore species?
Does the protection status of the PNMS affect key coastal-pelagic and nearshore species, including baitfish - how?
Can we monitor this over time?

Activity 2.2.3

4-6
years

Determine the impact of local, regional and global threats on pelagic fish populations

Activity indicator 2.2.3

By 2029, the most critical threats on pelagic fish populations within Palau have been identified, the impact of these characterised and quantified, locally significant management and regulatory guidance developed and key variables and approaches for long-term monitoring developed.

Activity products 2.2.3

1. A detailed list of key/most critical threats to pelagic fish populations.
2. Quantified impacts from these threats.
3. Locally significant management and regulatory guidance.
4. Key variables and approaches to track this over time.

Milestone questions 2.2.3

Do we know what threatens pelagic fish populations in Palau?
Do we know the scale - global, regional, local?
Can we do something about these threats and if so, what?
Can this be tracked over time?

Research objective 2.3

Define the regional role of the PNMS in supporting tuna populations.

Objective indicator 2.3

By 2034, sufficient data and information are available to determine the regional role of the PNMS in supporting tuna populations.

Context: Tuna is a major economic priority for Palau and the region, and this important marine resource is managed on a regional level. Research objective 2.3 aims to determine if locally distinct tuna populations exist and if there is a need for localized management. In addition, this objective aims to determine the value of the PNMS for tuna populations within the region.

Activity 2.3.1

Investigate the population structure of tuna in the region

1-3
years

Activity indicator 2.3.1

By 2026, the population structure of key tuna species within the WCPO are known, and implications for local management of tuna stocks understood.

Activity products 2.3.1

1. Data on the population structure of tuna populations in the region.
2. Management implications detailed and developed.

Milestone questions 2.3.1

Are there locally distinct tuna populations within Palau?
If locally distinct populations exist - what is their status and how should they be managed?
If so, does regional management address the needs of locally distinct populations?

Activity 2.3.2

Determine the conservation value of the PNMS for tuna populations

4-6
years

Activity indicator 2.3.2

By 2029, the value of the PNMS to conserve tuna populations has been characterised and quantified, and locally significant management and regulatory guidance developed, with key variables and approaches for long-term monitoring developed.

Activity products 2.3.2

1. Data on tuna utilization of the PNMS - spawning, nursery etc.
2. Determination of the ecological value of the PNMS to sustain tuna populations.
3. Locally significant management and regulatory guidance.
4. Key variables and approaches to track this over time.

Milestone questions 2.3.2

Is the PNMS an important spawning and nursery area for tuna?
How important is the PNMS for tuna populations - does the protection matter?
Can the protection value be quantified?
Can we track this over time?

Knowledge goal 3

Sustainable Development

In the next 10 years, Palau will have the knowledge it needs to take advantage of the contribution of the PNMS to human wellbeing.

Research objective 3.1

Document the social and cultural opportunities, values, benefits, costs and perceptions to inform decision making in the PNMS.

Objective indicator 3.1

By 2034, the social-cultural values, benefits, costs and perceptions of the PNMS within Palau have been identified, characterised and quantified.

Context: Large and remote marine protected areas have unique human dimension implications. The PNMS might be far removed from land and local communities, but this large-scale conservation action has important social, economic and cultural implications for Palau. Research objective 3.1 will uncover and dive into these implications.

Activity 3.1.1

Illuminate past and current cultural connections and knowledge of Palauan communities to pelagic and deep-sea environments

4-6
years

Activity indicator 3.1.1

By 2029, the cultural history of Palau's pelagic and deep-sea environments has been identified and characterised.

Activity products 3.1.1

1. Historic accounts of Palau's pelagic and deep-sea connections;
2. Locally significant policy and management guidance that highlight and support the cultural role of the PNMS

Milestone questions 3.1.1

Do we understand the cultural and historic connection with the pelagic and deep-sea environment?
Has an opportunity been given for communities to tell their stories?

Activity 3.1.2

Determine perceptions, understanding, and support of the PNMS among local residents and tourists

1-3
years

Activity indicator 3.1.2

By 2026, the understanding, level of support and perceptions of the PNMS by local communities within all 16 states have been quantified, with key variables and approaches for long-term monitoring developed.

Activity products 3.1.2

1. Data of community understanding, perceptions and support of the PNMS;
2. Locally significant management guidance that highlights and supports the human dimensions of the PNMS;
3. Key variables and approaches to track this over time.

Milestone questions 3.1.2

What do the people of Palau think of the PNMS?
Do the people of Palau support the PNMS - why or why not?
Can we track this over time?

Activity 3.1.3

Determine the direct and indirect social, cultural, and economic benefits and tradeoffs of the PNMS

4-6
years

Activity indicator 3.1.3

By 2029, the key social, cultural and economic benefits and tradeoffs of the PNMS for Palau and local communities across all 16 states have been identified, characterised and quantified, with key variables and approaches for long-term monitoring developed.

Activity products 3.1.3

1. Social and cultural data on the value of the PNMS;
2. Economic data on the value of the PNMS;
3. A documented understanding of the benefits and tradeoffs of the PNMS;
4. Key variables and approaches to track this over time.

Milestone questions 3.1.3

Are there any economic benefits from the PNMS - what are they?
How do Palauans benefit or not from the PNMS culturally and socially?
Can we track this over time?

Activity 3.1.4

Understand the local, national, and international political context surrounding the PNMS

7-10
years

Activity indicator 3.1.4

By 2034, major players and influences at the local, national, and international political context of the PNMS characterised and mapped.

Activity products 3.1.4

1. Geopolitical context of the PNMS characterised across various scales;
2. Geopolitical systems map.

Milestone questions 3.1.4

What is the geopolitical context of the PNMS?
How do international political negotiations, commitments, and trends impact the PNMS?
What policies/laws/conventions are at play?

Activity 3.1.5

Determine and track potential health benefits of the PNMS

1-3
years

Activity indicator 3.1.5

By 2026, healthy thresholds and nutritional benefits for consumption of key pelagic fish sourced in the PNMS has been determined, with key variables and approaches for long-term monitoring developed.

Activity products 3.1.5

1. Nutritional information for key pelagic fish species;
2. Health implications (with healthy thresholds) of a pelagic fish diet;
3. Key variables and approaches to track this over time.

Milestone questions 3.1.5

Do pelagic fish provide nutritional food - more so than other protein?
Are there any health implications from eating pelagic fish - what are they, what are the risks etc.?
What is the maximum recommended consumption of yellowfin, bigeye, and skipjack tuna, by group (e.g., women, children, etc.)?
Can we track this over time?

Research objective 3.2

Identify and quantify the ecosystem services and economic potential of the PNMS.

Objective indicator 3.2

By 2034, the key ecosystem services and the economic potential from the PNMS have been identified, characterised and quantified.

Context: Fisheries revenue is not the only economic contributor from the PNMS, and others are much less understood. Research objective 3.2 will determine and quantify other services provided by the PNMS.

Activity 3.2.1

Conduct a valuation of key ecosystem services on the economic potential of the PNMS to Palau

4-6
years

Activity indicator 3.2.1

By 2029, key ecosystem services provided by the PNMS have been identified, characterised and quantified, with key variables and approaches for long-term monitoring developed.

Activity products 3.2.1

1. Details of ecosystem services provided by the PNMS;
2. Monetary value of these services;
3. Key variables and approaches to track this over time.

Milestone questions 3.2.1

What ecosystem services are provided by the PNMS?
What is the monetary value of these services?
Can Palau take advantage of any of these services?
Can we track this over time?

Activity 3.2.2

Quantify the PNMS' economic contribution to GDP

1-3
years

Activity indicator 3.2.2

By 2026, the economic contribution of the PNMS towards Palau's GDP has been quantified, with key variables and approaches for long-term monitoring developed.

Activity products 3.2.2

1. Cost-benefit analyses of the PNMS;
2. Budgetary information on the contribution from the PNMS to Palau's GDP;
3. Variables and approaches for effective long-term monitoring;
4. A quantified estimate of the economic contribution to tourism from the PNMS.
- 5: Key variables and approaches to track this over time.

Milestone questions 3.2.2

Does the PNMS contribute to Palau's GDP, and if so, how?
How does the PNMS impact sector economics, including tourism, fisheries, conservation financing, etc.
Can we track this over time?

2.3 PNMS science and monitoring prioritisation and indicators

Research prioritisation

To ensure that this strategy provides a practical and priority driven roadmap for PNMS science and monitoring, the eight research objectives and 23 research activities underwent a prioritisation process. A PNMS science and monitoring prioritisation framework, adapted from the

Papahānaumokuākea Marine National Monument Natural Resources Science Plan (PNMN, 2011), was developed and each research objective and activity were prioritised accordingly. Once scored according to the prioritisation framework, each objective and activity was grouped into three different timeframes (short, medium and long term).

Prioritization Question 1

What are the risks to future protection, conservation and management of the PNMS of NOT obtaining this information in the next 5 years?



Prioritization Question 2

Will the research activity generate data relevant/useful to the PNMS goals (Healthy ocean populations, food security, sustainable development):

- ☐ All three (2) ☐ One or two (1) ☐ None (0)

Prioritization Question 3

Will this research produce data to help report on Palau's international commitments including, but not limited to, the Rio Conventions (i.e., CBD, UNCCD, UNFCCC) and the Sustainable Development Goals.

- ☐ Yes, to both (2) ☐ Yes, to one (1) ☐ No (0)

Prioritization Question 4

Will this research produce data useful in the management of the PNMS by Palau's government?

- ☐ Yes (1) ☐ No (0)

Prioritization Question 5

Is this research being done regionally/by a regional organization?

- ☐ Yes (1) ☐ No (0)

Research progress indicators

In the context of this strategy, research indicators are tools to help measure the implementation of the PNMS science and monitoring strategy. Indicators were developed following the SMART (specific, measurable, attainable, relevant, time bound) approach, to ensure that effective tracking of progress can take place. Long-term monitoring is critical to understand system change over time, and the effectiveness of conservation and management actions. To this end, the identification and establishment of the most suitable and effective long-term social, economic and ecological indicators for specific

research activities are incorporated within the indicators presented in this strategy. To help guide the outcomes for each research activity, proposed activity products are provided. Seeing that this strategy will undergo reviews as part of an adaptive management approach, milestone questions to ask during the review process are provided for each research activity. These questions will help assess the progress of research activities.

Scientific findings arising from research implemented in accordance with this strategy will contribute towards outcome indicators for the PNMS strategic plan 2023-2028. Outcome indicator needs have been fully integrated into this science and monitoring strategy.





2.4 Adaptive management

The research framework, prioritization methodology and indicator tracking efforts will be significant inputs into the approach towards adaptive management. Adaptive management processes are a path towards navigating the uncertainty inherent in managing natural systems. Through science-based information collection that is guided by management needs, the structured, iterative approach will provide for revisions to the science and monitoring strategy based on new knowledge or governance regimes. The content below describes the overarching philosophy for adapting the plan throughout the implementation timeline.

The core elements of adaptive management for the science and monitoring strategy will follow the procedures outlined in the PNMS strategic plan 2023-2028. These elements include outreach efforts and stakeholder engagement as mechanisms to gather new information. Special consideration regarding adjustments to the science and monitoring strategy may be triggered by a selection of forcing mechanisms, including:

● Activity Indicators Surpassing a Threshold

If stated indicators are not observably met or within reasonable achievement for the declared timeframe, then a review of the indicator, activity, and/or objective will be conducted.

● Event of Significant Environmental Consequence

If a natural or human-influenced event of significant consequence is observed (e.g., extreme warming; invasive species, increased storm frequency), then a review of the indicator, activity, objective and/or prioritization level will be conducted.

● Event of Significant Governance Consequence

If a legal, policy, or regulatory change to the governance structure (i.e., management, monitoring, outreach, enforcement) is formally approved and enacted, a determination will be made as to whether a review of the research framework is warranted.

● Periodic Review

If all indicators are meeting their stated thresholds within the proposed time frame, then a review of the research framework will be conducted in parallel with the review timeline stated in the PNMS strategic plan.

The determination of a confirmed adaptive management forcing mechanism will be made by the PNMS program manager in collaboration with lead representatives from PICRC and partner agencies.

2.5 Science and management

Results and products from the research activities outlined in this plan will have direct benefit to the people of Palau through a range of management and decision-making pathways. As delineated in the PNMS strategic plan 2023-2028, the government or semi-government agencies with direct responsibilities relevant to effective implementation of the Palau National Marine Sanctuary include an array of actors. The lists below provide the PNMS research community with additional context relevant to developing research projects that align directly with applied management decisions.

Two government and one semi-government agencies are mandated by law to play a role in management, administration, or enforcement of the PNMS. The table below outlines the agencies, the relevant divisions, the responsibility, and the decision pathway of interest to research efforts in the PNMS.



Agency	Division	Responsibility	Decision Pathway
MAFE Ministry of Agriculture, Fisheries, and the Environment	Bureau of Environment	Resource Use	Responsible for natural resource management and biodiversity, including international environmental convention and all protected species
	Bureau of Fisheries	Acts on behalf of the President to issue Research Permits in the PNMS	Alignment with international fisheries partnerships including Parties to the Nauru Agreement (PNA), Western and Central Pacific Fisheries Commission (WCPFC), Convention on Migratory Species (CMS) Also tasked with ensuring local food security through fisheries production
MOJ Ministry of Justice	Maritime Security	Law Enforcement	National-level enforcement of maritime boundaries and laws Alignment with international agreements including the United Nations Convention on the Law of the Sea (UNCLOS)
PICRC Palau International Coral Reef Center	PNMS Office	Administrator through coordination of research, education, outreach	Stakeholder engagement, including education and outreach Coordination of research activities

In addition to the agencies mandated to engage in implementing the PNMS, several related government agencies also hold decision making pathways of interest to the broader research community.

Agency	Decision Pathway
Ministry of Health and Human Services (MHHS)	Nutrition and food security guidance
Ministry of Human Resources, Culture, Tourism, and Development (MHRCTD)	Social and cultural ties Tourism impacts and benefits
Ministry of Finance (MOF)	Management of the Environmental Impact Fee Report to Olbiil Era Kelulau (OEK) on permitting in the Commercial fishing zone Spatial data through the Palau Automated Land and Resource Information System (PALARIS) office
Office of the President	Reporting on and alignment with global initiatives including the Sustainable Development Goals, United Nations Framework Convention on Climate Change, and the Convention on Biological Diversity Legal Authority to issue Research Permits in the PNMS
Olbiil Era Kelulau (OEK)	Engagement on legislative matters
Palau Visitors Authority (PVA)	Tourism impacts and benefits

The agencies and pathways noted above are subject to revision based on updated coordination agreements and amendments to the governance structure. Additional details regarding the direct avenues for engagement are stated in the PNMS strategic plan 2023-2028.

Section 3

References

Andres, M., Musgrave, R.C., Rudnick, D.L., Zeiden, K., Peacock, T., Park, J-H. 2020. On the Predictability of Sea Surface Height around Palau. *American Meteorological Society*. 50: 3267- 3294.

Andrews, O. 2013. Palau Marine Mammal Project Report. 33 pp.

Brown, C.J., Desbiens, A., Campbell, M.D., Game, E.T., Gilman, E., Hamilton, R.J., Heberer, C., Itano, D., Pollock, K. 2021. Electronic monitoring for improved accountability in western Pacific tuna longline fisheries. *Marine Policy*. 132: 104664.

Cimino, M.A., Anderson, M., Schramek, T., Merrifield, S., Terrill, E.J. 2019. Towards a Fishing Pressure Prediction System for a Western Pacific EEZ. *Nature: Scientific Reports*. 9(461): 1-10.

Claassens, L., Phillips, B., Ebert, D.A., Delaney, D., Henning, B., Nestor, V., Ililau, A., Giddens, J.L. Under Review. First records of the Pacific Sleeper shark *Somniosus cf. pacificus* in the western tropical Pacific. *Journal of Fish Biology*.

Dacks, R., Lewis, S.A., James, P.A.S., Marino, L.L., Oleson, K.L.L. 2020. Documenting baseline value chains of Palau's nearshore and offshore fisheries prior to implementing a large-scale marine protected area. *Marine Policy*. 117: 103754.

Fairbanks, L., Gruby, R. Unpublished data. Human Dimensions of Large Marine Protected Areas: A case study of the Palau National Marine Sanctuary.

Filous, A., Friedlander, A.M., Griffin, L., Lennox, R.J., Danylchuk, A.J., Mereb, G., Golbuu, Y. 2020. Movements of juvenile yellowfin tuna (*Thunnus albacares*) within the coastal FAD network adjacent to the Palau National Marine Sanctuary: Implications for local fisheries development. *Fisheries Research*. 230: 105688.

Filous, A., Friedlander, A.M., Toribiong, M., Lennox, R.J., Mereb, G., Golbuu, Y. 2022. The movements of yellowfin tuna, blue marlin, and sailfish within the Palau National Marine Sanctuary and the western Pacific Ocean. *ICES Journal of Marine Science*. 1-12.

- Fine, R.A., Lukas, R., Bingham, F., Warnar, M., Gammon, R. 1994. The Western Equatorial Pacific: A water mass crossroads. *Journal of Geophysical Research*. 99: 25063-25080.
- Friedlander, A.M., Golbuu, Y., Caselle, J.E., Ballesteros, E., Letessier, T.B., Meeuwig, J.J., Gouezo, M., Olsudong, D., Turchik, A., Sala, E. 2014. Marine biodiversity and protected areas in Palau: Scientific report to the government of the Republic of Palau. 47 pp.
- Gilman, E., Chaloupka, M., Merrifield, M., Malsol, N.D., Cook, C. 2016. Standardized catch and survival rates, and effect of a ban on shark retention, Palau pelagic longline fishery. *Aquatic Conservation: Marine and Freshwater Ecosystems*. 26(6): 1031-1062.
- Gruby, R.L., Fairbanks, L., Acton, L., Artis, E., Campbell, L.M., Gray, N.J., Mitchell, L., Zigler, S.B.J., Wilson, K. 2017. Conceptualizing Social Outcomes of Large Marine Protected Areas. *Coastal Management*. 45(6): 416-435.
- Gruby, R.L., Gray, N.J., Fairbanks, L., Havice, E., Campbell, L.M., Friedlander, A., Oleson, K.L.L., Sam, K., Mitchell, L., Hanich, Q. Policy interactions in large-scale marine protected areas. *Conservation Letters*. 14(1): e12753.
- Hari, K., Jaiteh, V., Chin, A. 2021. The sharks and rays of Palau: biological diversity, status, and social and cultural dimensions. *Pacific Conservation Biology*. 28(5): 398-413.
- Heron, S.F., Metzger, E.J., Skirving, W.J. 2006. Seasonal Variations of the Ocean Surface Circulation in the Vicinity of Palau. *Journal of Oceanography*. 62: 413-426.
- Jaiteh, V., Peatman, T., Lindfield, S., Gilman, E., Nicol, S. 2021. Bycatch Estimates from a Pacific Tuna Longline Fishery Provide a Baseline for Understanding the Long-Term Benefits of a Large, Blue Water Marine Sanctuary. *Frontiers in Marine Science*. 8:720603.
- Johnston, T.M.S., Schonau, M.C., Paluszkiwicz, T., MacKinnon, J.A., Arbic, B.K., Colin, P.L., Alford, M.H., Andres, M., Centurioni, L., Graber, H.C., Helfrich, K.R., Hormann, V., Lermusiaux, P.F.J. Musgrave, R.C., Powell, B.S., Qiu, B., Rudnick, D.L., Simmons, H.L., St. Laurent, L., Terrill, E.J., Trossman, D.S., Voet, G., Wijesekera, H.W., Zeiden, K.L. 2019. Flow Encountering Abrupt Topography (FLEAT) - A multiscale observational and modelling program to understand how topography affects flows in the western north pacific. *Oceanography*. 32(4): 10-21.
- Klain, S., Eberdong, J., Kitalong, A., Yalap, Y., Matthews, E., Eledui, A., Morris, M., Andrew, W., Albis, D., Kemesong, P. 2007. Linking Micronesia and Southeast Asia: Palau Sea Turtle Satellite Tracking and Flipper Tag Returns. *Marine Turtle Newsletter*. 118: 9-11.
- Kobayashi, K. 2004. Origin of the Palau and Yap trench-arc systems. *Geophysical Journal International*. 157: 1303-1315.
- Kobayashi, K., Fujioka, K., Fujiwara, T., Iwabuchi, Y., Kitazato, H. 1997. Why is the Palau Trench so Deep? Deep-Sea Trench without Plate Convergence. *Proceeding of the Japan Academy*. 73(B): 89-94.
- Koshiha, S., Wongbusarakum, S., Marino, L.L., Tellei, I., Uchel, A., Jonathan, R., Claassens, L., Bukurou, L., Olsudong, D., Bukurrou, A., Gorstein, M. 2022. Understanding the perceived human well-being benefits and values of the Palau National Marine Sanctuary: A Pilot Study in Airai State. PICRC Technical Report No. 22-02. 33 pp.
- Lemenkova, P. 2019. Automatic Data Processing for Visualising Yap and Palau Trenches by Generic Mapping Tools. *Cartographic Letters, Cartographic Society of the Slovak Republic*. 27(2): 72-89.
- Lewis, S.A., Fezzi, C., Dacks, R., Ferrini, S., James, P.A.S., Marino, L.L., Golbuu, Y., Oleson, K.L.L. 2020. Conservation policies informed by food system feedbacks can avoid unintended consequences. *Nature Food*. 1: 783-786.
- Li, Y., Song, L., Nishida, T., Gao, P. 2012. Development of integrated habitat indices for bigeye tuna, *Thunnus obesus*, in waters near Palau. *Marine and Freshwater Research*. 63: 1244-1254.
- McPhaden, M.J., Picaut, J. 1990. El Niño-Southern Oscillation Displacements of the Western Equatorial

Pacific Warm Pool. *Science*. 250(4986): 1385-1388.

Meyers, M.M., Francis, M.P., Erdmann, M., Constantine, R., Sianipar, A. 2020. Movement patterns of whale sharks in Cenderawasih Bay, Indonesia, revealed through long-term satellite tagging. *Pacific Conservation Biology*. 26(4): 353-364.

Micronesia Challenge. 2006. MC Final Report: Micronesia Challenge Action Planning Meeting, December 4-7, 2006. 12 pp.

Musyl, M.K., Gilman, E.L. 2018. Post-release fishing mortality of blue (*Prionace glauca*) and silky shark (*Carcharhinus falciformes*) from a Palauan based commercial longline fishery. *Reviews in Fish Biology and Fisheries*. 28: 567–586.

Palau Conservation Society (PCS). 2023. Palau National Marine Strategy Strategic Plan 2023-2028, Volumes 1 and 2. Koror, Palau: PICRC.

Palau International Coral Reef Center and Stanford Center for Ocean Solutions, “Palau’s National Marine Sanctuary: Managing Ocean Change and Supporting Food Security,” PICRC, December 2019. 80 pp. Available at: <https://picrc.org/picrcpage/palau-national-marine-sanctuary/> and <https://oceansolutions.stanford.edu/pnms-report>.

Palau National Marine Sanctuary Office. 2017. Palau National Marine Sanctuary 5-year strategic plan: Sustaining Palau’s oceanic resources for environmental and social wellbeing. Koror, Palau. 22 pp.

Qiu, B., Chen, S., Powell, B.S., Colin, P.L., Rudnick, D.L., Schonau, M.C. 2019. Nonlinear short-term upper ocean circulation variability in the tropical western Pacific. *Oceanography*. 23-31.

Papahānaumokuākea Marine National Monument. 2011. Papahānaumokuākea Marine National Monument Natural Resources Science Plan. Hawaii. 88 pp.

RPPL 10-35. 2019. Palau National Marine Sanctuary Act Amendment. <https://www.palau.gov/wp-content/uploads/2019/07/RPPL-No.-10-35-re.-Amendment-to-the-PNMS.pdf>.

RPPL 9-49. 2015. Palau National Marine Sanctuary Act. <http://palau.gov/wp-content/uploads/2015/10/RPPL-No.-9-49-Palau-National-Marine-Sanctuary-Act.pdf>.

Schönau, M.C., D.L. Rudnick. 2015. Glider observations of the North Equatorial Current in the western tropical Pacific. *Journal of Geophysical Research*. 120(3): 586–605.

Schonau, M.C., Wijesekera, W., Teague, W.J., Colin, P.L., Gopalakrishnan, G., Rudnick, D.L., Cornuelle, B.D., Hallock, Z.R., Wang, D.W. 2019. The End of an El Niño – A view from Palau. *Oceanography*. 32(4): 33-45.

Tanabe, T., Ogura, M., Takahashi, M. 2001. Early life ecology of skipjack tuna, *Katsuwonus pelamis*, based on survey cruises off Palau and Micronesia, 1992-1996. *Japan Agricultural Research Quarterly*. 35(1): 67-77.

Villasenor-Derbez, J.C., Lynham, J., Costello, C. 2020. Environmental market design for large-scale marine conservation. *Nature Sustainability*. 3: 234–240.

Yesson, C., Clark, M.R., Taylor, M., Rogers, A.D. 2011. The global distribution of seamounts based on 30-second bathymetry data. *Deep Sea Research Part I: Oceanographic Research Papers* 58: 442-453.

A black and white photograph of an underwater scene. In the foreground, a large, flat, silver fish (likely a pomfret) swims towards the left. Behind it, several other fish of various species are visible, including a larger, more elongated fish with a prominent dorsal fin. The background features the dark, vertical structure of a ship's mast or rigging, with many thin, dark lines (ropes or wires) extending from it. The water is dark and slightly murky, with some light reflecting off the fish's scales and the ship's structure.

Section 4

Appendices

4.1 PNMS science and monitoring expert workshop attendees

Name	Agency / Company	Country
Fiorenza Micheli	Oceans Department and Stanford Center for Ocean Solutions, Stanford University	USA
Eric Hartge	Stanford Center for Ocean Solutions	USA
Collin Closek	Stanford Center for Ocean Solutions	USA
Natalie Arnoldi	Stanford University	USA
Kirsten Oleson	University of Hawaii	USA
Robert Richmond	University of Hawaii	USA
Constance Rambourg	University of Hawaii	USA
John Lynham	University of Hawaii	USA
Randi Rotjan	Boston University	USA
Rebecca Gruby	Colorado State University	USA
Megan Cimino	University of California Santa Cruz	USA
Chrissy Hernandez	Cornell University	USA
Quentin Hanich	University of Wollongong	Australia
Jon Day	James Cook University	Australia
Daniel Wagner	Ocean Exploration Trust	USA
Eric Terrill	Scripps Institution of Oceanography	USA
Sophia Merrifield	Scripps Institution of Oceanography	USA
Caitlin Adams	NOAA Expedition & Exploration Division	USA
Amanda Netburn	NOAA Office of Ocean Exploration and Research	USA
Jeremy Weirich	NOAA Ocean Exploration	USA
Kevin Mackay	South and West Pacific Data Center, Seabed 2030. National Institute of Water and Atmospheric Research (NIWA)	New Zealand
Iwao Ueki	Japan Agency for Marine-Earth Science and Technology (JAMSTEC)	Japan
Atuatasi Lelei Peau	National Marine Sanctuary of American Samoa	American Samoa
Genevieve Brighthouse	National Marine Sanctuary of American Samoa	American Samoa
Isabel Halatuituia	National Marine Sanctuary of American Samoa	American Samoa
Valerie Brown	National Marine Sanctuary of American Samoa	American Samoa
Xavier E. Matsutaro	Office of Climate Change, Bureau of Budget & Planning, Ministry of Finance	Palau
Kiblas Soaladaob	Bureau of Cultural and Historical Preservation, Ministry of Human Resources, Culture, Tourism and Development	Palau
Jeremiah Ngiratreged	Office of the Vice President, Ministry of Justice	Palau
Ophelia Ongalibang	Office of the Palau Automated Land and Resources Information System (PALARIS)	Palau
Patrick L. Colin	Coral Reef Research Foundation	Palau
Alex Filous	The Nature Conservancy	Palau
Sherry Constantine	The Nature Conservancy	Palau
Okada Techitong	Chairman of Belau Offshore Fishers Incorporated (BOFI)	Palau
Louw Claassens	Palau International Coral Reef Center (PICRC)	Palau
Yimnang Golbuu	Palau International Coral Reef Center (PICRC)	Palau
Geraldine Rengiil	Palau International Coral Reef Center (PICRC)	Palau
King Sam	Palau International Coral Reef Center (PICRC)	Palau
Elsei Tellei	Palau International Coral Reef Center (PICRC)	Palau

4.2 Research permitting guidance and key contact information

All research conducted within the Palau National Marine Sanctuary requires a national permit from the Government of Palau. Permit applications should be submitted to the Bureau of Fisheries, Ministry of Agriculture, Fisheries and Environment (MAFE). Additional research permits may be required from the Bureau of Cultural and Historical Preservation or other agencies. Proposed research projects should be discussed with MAFE prior to preparation and submission of a permit application. The application should include an application form, detailed project proposal and payment of a permit fee, paid directly to the Ministry of Finance.

If approved, permits are issued with project-specific conditions which must be complied with, including data sharing requirements. International convention requirements, such as Access and Benefit Sharing of genetic material, may also apply.

If research is conducted within the jurisdiction of a state (up to 12 NM from the shore), a state permit is required in addition to a national permit. State governments should be contacted for state permit requirements (see state contact information below), which varies across individual states.

Key contact information:

	Tel	E-mail	Website
Ministry of Agriculture, Fisheries and Environment	(680) 767-3125	palau.mafe@palaugov.org	https://www.palaugov.pw/executive-branch/ministries/agriculture-fisheries-and-environment/
PALARIS	(680) 488-6838/6654 or 767-8638	palaris@palaunet.com	
Ministry of Human Resources, Culture, Tourism & Development	(680) 488-6600	info@ministryofhrctd.pw	https://ministryofhrctd.pw/ Palau Historic Research Library: https://airtable.com/shrCBruOm3fJWOrf
Division of Maritime Security and Fish & Wildlife Protection	(680) 488-5206		https://www.palaugov.pw/executive-branch/ministries/justice/bureau-of-public-safety/division-of-maritime-security-and-fish-wildlife-protection/
Palau International Coral Reef Center	(680) 488-6950	picrc@picrc.org	https://picrc.org/
Coral Reef Research Foundation	(680) 488-5255	crrfpalau@gmail.com	https://coralreefpalau.org/
Palau Conservation Society	(680) 488-3993	admin@palaupreservation.org	https://www.palaupreservation.org/
Ebiil Society		info@ebiilsociety.org	https://www.ebiilsociety.org/
Friends of the PNMS		friendsofnpms@gmail.com	https://www.palaupledge.com/

	Tel	E-mail	Website
One Reef	(680) 488-3993	info@onereef.org	https://onereef.org/
Micronesia Challenge		micronesia.challenge2030@gmail.com	http://themicronesiachallenge.blogspot.com/
Belau Offshore Fishing Incorporation	(680) 488-5093	bofipalau@gmail.com	https://www.facebook.com/belauoffshorefishersinc/
Palau Commercial Fishing Company	(680) 775-8672	palaucommercialfishingco@gmail.com	https://www.facebook.com/profile.php?id=100086719726393

Palau state offices

	Tel	E-mail
Koror State	(680) 488-2439	kororstategov@gmail.com
Airai State	(680) 587-3511	airaigov@gmail.com dnabo@airaistategov.com governor.norman@airaistategov.com
Ngatpang State	(680) 535-1882	ngatpang.nsg3@gmail.com
Ngeremlengui State	(680) 733-2967	ngeremlenguistategovt@gmail.com
Ngardmau State	(680) 747-2967	ngardmaustate@gmail.com
Ngaraard State	(680) 824-4490	ngaraard18@gmail.com
Ngarchelong State	(680) 855-2967 (680) 488-2871	nsg@palaugov.org
Kayangel State	(680) 876-2967 (680) 488-2766	kayangel2766@gmail.com
Ngiwal State	(680) 679-2967	ngiwalstate79@gmail.com
Melekeok State	(680) 654-2967	melekeokgov@gmail.com
Ngchesar State	(680) 622-2967 (680) 488-5881	ngchesarstategovernment@gmail.com
Peleliu State	(680) 488-5282 (680) 277-2967	peleliustategov@gmail.com
Angaur State	(680) 488-2439	angaurstate@gmail.com
Sonsorol State	(680) 488-1237	sonsorol@palaunet.com
Hatohobei State	(680) 488-2218	hatohobei2022@gmail.com
Aimeliik State	(680) 544-2967	asgov544@gmail.com

4.3 Linkages between the PNMS science and monitoring strategy 2024-2034 and the PNMS strategic plan 2023-2028

Science and Monitoring Activities		Knowledge Goals									
		3: Sustainable Development 2: Healthy ocean populations & ecosystems 1: Food Security									
		3.2.2: PNMS economic contribution to GDP 3.2.1: Valuation of ecosystem services 3.1.5: Health benefits of the PNMS 3.1.4: Local, national, economic benefits & tradeoffs 3.1.3: Social, cultural, international political context 3.1.2: Resident & tourist perceptions, understanding 3.1.1: Past & current cultural connections 2.3.2: Conservation value of PNMS for tuna 2.3.1: Population structure of tuna in the region 2.2.3: Local, regional & global threats & impacts 2.2.2: PNMS & coastal-pelagic & nearshore fish 2.2.1: Spatial & temporal dynamics 2.1.2: Viability of DFZ & AFZ for local biodiversity 2.1.1: Fishery supply & demand dynamics 1.3.2: Conservation of global to local biodiversity 1.3.1: Conservation value of PNMS for species 1.2.4: Adaptive capacity for climate scenarios 1.2.3: Climate-change induced losses & damages 1.2.2: Document & measure climate change impacts 1.1.1: Synthesize information & Gap Analysis									
PNMS Strategic Plan Information Needs (by Goal)											
GOAL 1: ENABLING ENVIRONMENT AND COORDINATION:		Strengthened CAPACITY AND ABILITY of core partner institutions to responsibly steward the PNMS and deliver on their related mandates, in an increasingly improved and supportive ENABLING ENVIRONMENT.									
Institutional arrangements for optimal effectiveness											
Marine spatial features											
Management effectiveness and capacity measurements											
Communications arrangements for optimal effectiveness											
Sustainable financing needs (now and future)											
Information needs and gaps											
National and individual incomes and financial benefits and projections (now and over time)											
Supply chain links and investments											
GOAL 2: FOOD SECURITY:		Increased FOOD SECURITY for the Palauan people, both from increasing production, access, and affordability of PNMS-sourced proteins and from increased economic gains for government, businesses, and individuals.									
Fishery production and sustainability											
Fishery supply and value chain											
Local and international fishery projections and harvests											
Tuna fishery parameters											
FAD fishery parameters											
Baitfish parameters and production											

																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								</
--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	----

